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The Editor
Journal of Pharmaceutical Innovation

Submission of a Research Article: “An Evidence-Graded Framework for Pharmaceutical Route Selection and Process-Risk Prioritization: A PN6047 Case Study”

Dear Editor,

I submit the enclosed paper for consideration in the Journal of Pharmaceutical Innovation as a Research Article. It addresses a question that arises before any process chemistry is run on a new drug substance: how can the public information on a discovery-scale route be converted into an auditable plan for process development? The principal contribution is not a proposed synthesis of PN6047, but a reproducible decision framework for converting incomplete discovery-route information into evidence grades, process-development risks, sensitivity-tested priorities, and a stage-gated experimental validation plan. PN6047, a clinical-stage δ -opioid receptor agonist whose route is public only as milligram-to-gram patent examples, is the case study.

What the paper does

- It inventories the operations exemplified in WO 2016/099393 and WO 2022/013153 and grades each transformation, and its conditions, in four evidence classes (direct, close precedent, general precedent, hypothesis), so that what is established, inferred and proposed is never mixed.
- It flags documented regulatory and hazard attributes of the exemplified route (11 flag instances, including an ICH Q3C Class 1 solvent, palladium, a benzotriazole-derived reagent and preparative HPLC) and screens two process-oriented variants: they carry 3 flags each, but their validation burden is higher (11 and 15 against 2 on the linear scale), because their conditions are untested: fewer documented flags did not mean lower development uncertainty.
- It tests the sensitivity of the prioritisation of a twelve-item risk register to rating uncertainty, conditional on the initial risk model; two risks, impurity purge when crystallisation replaces chromatography and the undisclosed upstream route to the vinyl bromide building block, lead under three rating distributions.
- It ends with a stage-gated experimental programme, and states what is not claimed: no synthesis was performed, and no yield, impurity level, cost or timeline is predicted.

Why the journal

The study concerns route selection, process risk, impurity control, starting-material justification (ICH Q11) and the evidence needed before scale-up, which I believe fall within the journal's coverage of manufacturing and applied pharmaceutical science, and the framework can be applied to other compounds whose discovery routes are public. It follows the Research Article format and the abstract, summary and teaser requirements.

Data, code and disclosures

All inputs, model code, tests and figure scripts are public at <https://github.com/sandlerleon/pn6047-route-design> and archived at <https://doi.org/10.5281/zenodo.23200725>; the manuscript is available as a preprint at <https://doi.org/10.5281/zenodo.23201405>. A large language model (Claude, Anthropic) was used to assist with retrieval, code, figures and drafting; I reviewed and verified the content, resolved every reference through Crossref, and take full responsibility, as stated in Section 2.10. I am the sole author, have no competing interests, received no funding and have no affiliation with the applicant of the patents analysed. An earlier, shorter and unpublished outline of the route concept was submitted to an open-innovation challenge, as disclosed in the Declarations. The manuscript has not been published and is not under consideration by any other journal.

Thank you for considering the paper.

Yours sincerely,

Leon Sandler